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On the Relative Importance of Vocal Source, System, and Prosody in Human Depression

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Abstract¹

In Major Depressive Disorder (MDD), neurophysiologic changes can alter motor control [1][2] and therefore alter speech production by influencing vocal fold motion (source), the vocal tract (system), and melody (prosody). In this paper, we use a database of voice recordings from 28 depressed subjects treated over a 6-week period [3] to compare correlations between features from each of the three speech-production components and clinical assessments of MDD. Toward biomarkers for audio-based continuous monitoring of depression severity, we explore the contextual dependence of these correlations with free-response and read speech, and show tradeoffs across categories of features in these two example contexts. Likewise, we also investigate the context- and speech component-dependence of correlations between our vocal features and assessment of individual symptoms of MDD (e.g., depressed mood, agitation, energy). Finally, motivated by our initial findings, we describe how context may be useful in “on-body” monitoring of MDD to facilitate identification of depression and evaluation of its treatment.

Index Terms: major depressive disorder, motor control, clinical assessment, vocal biomarkers, free-response vs. read speech

1.0 INTRODUCTION

MAJOR DEPRESSIVE DISORDER (MDD) is the most prevalent mood disorder, with a lifetime risk observed to fall between 10-20% for women and 5-12% for men [4]. As the number of people suffering from MDD steadily increases, so too does the burden of accurate diagnosis. Currently, diagnosis of MDD requires clinical assessment by a professional with significant clinical experience. Additionally, the specific scores assigned to monitor the depression of the patient, will have a level of inter-clinician variability, making the tracking of medication efficacy in a clinical trial challenging. The growing global burden of MDD suggests that a convenient and automated method to assess depression severity would both simplify and standardize the task of diagnosing and monitoring depression, and allow for greater availability and uniformity in assessing

depression. One approach that avoids in-office clinical visits, and thus may facilitate automated measurement and identification, as well as quicken evaluation of treatment, uses “on-body” sensor-based platforms to extract signals to be used as biomarkers.

A class of potential biomarkers of growing interest is based on vocal characteristics, which have been shown to change with a patient’s mental condition and emotional state [3][6-12]. Such changes reflect the perception of qualities such as monotony, slur, hoarseness, and breathiness in voices of depressed individuals. In this paper, these aspects of the depressed voice can be considered – perhaps loosely and by no means exclusively – to be associated with one of three speech production components (Figure 1): the vocal tract (system), its excitation (source), and overriding melody (prosody). Monotony may be associated with prosodic (e.g., modulation of pitch, energy, and speech rate) and slur with system (dynamics of vocal tract articulators) characteristics. Hoarseness, breathiness, and glottalization may be associated with source characteristics (vocal fold motion and level of aspiration).

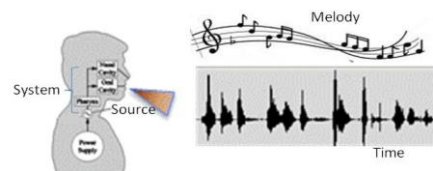


Figure 1: Schematic of speech source (at the vocal folds), system (vocal tract), and melody (prosody)

Although not completely consistent, studies have shown vocal features to correlate with depression, and have been incorporated into the development of automated classifiers for detecting depression [3][6-12]. Nevertheless, the relative importance of these features as categories of source, system, and prosody in different contexts has not been addressed, nor has there been a study of how these categories and contexts relate to the individual symptoms of MDD that comprise the total assessment of MDD severity. These considerations are important for our longer-term objective of continuous monitoring in different life scenarios.

In this work, we posit that the effect of depression on different aspects of motor control is reflected in speech components in diverse and measureable ways. Furthermore, we examine these potential differences in the context of open question (free response) speech, and speech produced in reading a standard passage. Since some of the brain regions activated by free vs. read speech differ [13], we surmise that speech components extracted from these contexts differ in their correlation patterns with assessed MDD symptoms.

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To explore these hypotheses, we use a 28-subject database of voice recordings from depressed patients treated over a 6-week period in an open-label, naturalistic study design collected by Mundt et al. [3]. This includes scores from two standard clinical rating scales to assess depression severity: the 17-item Hamilton Depression Rating Scale (HAM-D), administered by a trained clinician, and the self-reported Quick Inventory of Depressive Symptomatology (QIDS), completed electronically by the patient. Each scale provides a total depression severity score, as well as the ratings of the individual symptoms of depression comprising the scales. Examples of individual depressive symptoms rated by the scales include psychomotor retardation, anxiety, agitation, and energy.

For the 28-subject dataset used in this study, we observe that features extracted from the three components of speech show different patterns of correlation with total and individual-symptom assessment depression severity. The correlation of an individual symptom of MDD changes with each feature, can be higher than that of the total assessment score, and can show sizable difference between free-flow and read speech. Moreover, these correlations exhibit some consistency within a speech component category (source, system, prosody).

Our paper is organized as follows. In Section 2, we describe the 28-subject MDD database collected by Mundt et al. [3]. In Section 3, we describe our signal-processing methodologies for vocal-feature extraction for each speech-component category. In Sections 4 and 5 we describe results of our correlation analyses, and in Section 6 provide conclusions and projections toward future work.

2.0 DEPRESSION DATABASE

2.1 Database

We analyze voice data originally collected by Mundt et al. [3] in a depression-severity study that includes telephone-based speech recordings collected from patients during office visits. Thirty-five physician-referred subjects (20 women and 15 men, mean age 41.8 years) participated in the study. All of the subjects had recently started on pharmacotherapy and/or psychotherapy for depression, and continued treatment over the 6-week assessment period of the study. The speech data consists of recordings (sampled at 8 kHz) over a standard office telephone at weeks 0, 2, 4, and 6. Subjects also completed the self-reported QIDS using a validated interactive voice response (IVR) system [3]. To avoid confounds introduced by the use of different telephones, we use only speech samples of conversational (free-response) speech, the read “Grandfather” passage, and distinct vowels (/a/, /ae/, /i/, /u/) recorded while the subjects were present in the clinic and speaking into the local telephone. In addition, our analyses include only data from subjects who completed the entire longitudinal study, resulting in analysis of 28 of the original 35 patients. Approximately 3-6 minutes of speech per session (i.e., per day) were available for analysis. The free-response and Grandfather passage are used for vocal tract and prosody analysis, while the held vowels are used for the source analysis.

For this database, the standard method of evaluating depression severity was the clinician-rated 17-item HAM-D assessment [3][24]. Total HAM-D scores reflect the summation of the 17 rated symptoms of MDD, each rated on a scale of (0-2) or (0-4). The self-reported QIDS assesses 16 symptoms of depression, which are used to derive 9 depression domain scores (scored 0-3) corresponding to the DSM-IV diagnostic criteria for MDD [4]. Table 1 gives the

	CLN/IVR-HAM-D Sub-symptom	QIDS Sub-symptom
1	Depressed mood	Sleep
2	Guilt	Mood
3	Suicide	Appetite and/or weight change
4	Initial insomnia	Concentration
5	Middle insomnia	Outlook
6	Terminal insomnia	Suicide
7	Work and interests	Involvement
8	Psychomotor retardation	Energy
9	Psychomotor agitation	Max of psychomotor agitation/retardation
10	Psychological anxiety	
11	Somatic anxiety	
12	Appetite	
13	Somatic symptoms general	
14	Genital symptoms	
15	Hypochondriasis	
16	Weight loss	
17	Insight	

Table 1: MDD symptoms rated in the HAM-D and QIDS scales.

individual MDD symptoms assessed by each scale.

2.2 Previous results on Mundt et al. database

A study by Mundt et al. [3] investigated correlations of numerous parameters related to pitch and speech rate with HAM-D total scores (no individual symptom ratings). Using average speech rate (syllables/sec), they found a Pearson correlation of $r = -0.23$ ($p < 0.001$). Mundt et al also found some correlation of the variance of the second formant with HAMD assessment (-0.17 for a $p < 0.05$) but no corresponding correlation for the first formant (the two formants studied). In a second related work using the Mundt et al. database, Trevino et al. [11] dissected average measures of speech rate into phone-specific characteristics. Combined phone-duration measures uncovered stronger relationships between speech rate and depression severity than the global measures previously reported as speech-rate biomarkers. Trevino et al. also developed a general framework for analyzing individual symptom categories through phonological units, and showed for the first time that a biomarker (e.g., speech rate) can correlate with sub-symptoms (e.g., psychomotor retardation) more highly than with a total assessment, a result that motivated the symptom-specific approach in this paper. Nevertheless, in the current work we use average speech rates. Other previous work on the Mundt et al. database has reported preliminary correlation and classification results using mel-cepstral [12] and source [7] features.

3.0 SPEECH PRODUCTION COMPONENTS

Our features fall into three categories (Figure 1): source (at the vocal folds), system (the vocal tract), and melody (prosody). The source relates to motion of the vocal folds, which can vibrate to provide pseudo-periodic air pulses (during “voiced” speech) as input to the vocal tract, and sets the “pitch”, which is the fundamental frequency of vibration. This input is spectrally shaped by the vocal tract resonant chamber (system). The speech melody (prosody) is characterized by the modulation of pitch, energy, and rate of speaking. We selected

a preliminary set of vocal features that reflect change in the dynamics of speech production that may occur with a change in motor control in the depressed state.

3.1 Source

To characterize the source, we use vocal-fold jitter and shimmer, which are variations in the amplitude and timing of vocal-fold vibrations, respectively, and aspiration, which is turbulent noise at the folds. The extent of these variations may reflect a change in motor control by the vocal folds. Due to limitations of state-of-the-art algorithms on running speech, corresponding features are calculated on held vowels only.

Jitter: Jitter is the period-to-period variation in pulse timing during voicing. To estimate this quantity, we extract the approximate source pulse times for voiced regions using the speech-signal processing tool Praat [14]. A jitter feature is then calculated as the average absolute difference between consecutive intervals, divided by the average length, and multiplied by 100 to yield a percent value.

Shimmer: Shimmer measures the period-to-period variation in source pulse amplitude in voiced regions. To measure this quantity, we use the Praat tool to first extract approximate source pulse times for voiced regions, and then compute amplitudes at the pulse times. Our shimmer feature is the average absolute difference between the amplitudes of consecutive periods, divided by the average amplitude, and multiplied by 100 to yield a percent value.

Aspiration: To measure aspiration, we use a harmonic/noise decomposition technique referred to as *pitch-scaled harmonic filtering* (PSHF) [16]. The PSHF approach uses an analysis window duration equal to four pitch periods and relies on the property that harmonics of the fundamental frequency fall at specific frequency bins of the short-time Fourier transform. We then perform spectral subtraction to obtain the noise component spectrum. Our pitch periods are obtained using the Praat tool. This decomposition technique approximately isolates the noise component in an aspirated utterance. The resulting voiced and aspiration components form a harmonics-to-noise ratio (HNR).

3.2 System

We use articulation, herein loosely associated with formant dynamics, as a means of understanding changes in the system speech component. There are a variety of approaches to the on-going challenge of formant estimation and tracking. We have selected an algorithm recently developed by Rudoy, Mehta, Spendley, and Wolfe [17][18], based on the principle that formants are correlated with one another in both the frequency and time domains. Embedded in the algorithm is a voice-activity detector that allows a Kalman predictor to smoothly coast consistently through non-speech regions.

Our formant features are given by: formant variance, the mean of formant velocity, and the mean of formant acceleration, with the average of the first forward and backward differences across three consecutive frames used to compute derivatives (as performed throughout the paper for velocity and acceleration measurements). These features are taken across analysis frames (10 ms) relative to a running mean derived by a 50-Hz-cutoff lowpass filter, and provide estimates of vocal tract formant dynamics around low-frequency target trajectories. We hypothesize these formant dynamics reflect physiological control of vocal tract trajectories around phonetic targets, and thus are correlated with motor symptoms of depression.

3.3 Prosody

We characterize prosody as the variability and dynamics of pitch and energy, and average speech rate.

Pitch: We investigate three different measures of pitch: pitch variance, and mean and variance of pitch velocity. A sinusoidal-based algorithm is used for forming our pitch estimates [20], which are taken every 10 ms in voiced regions.

Energy: We investigate four different measures of energy: the mean and variance of energy and energy velocity. Energy and velocity are computed at 10-ms frames. Energy is measured in only active speech frames, which are determined according to an energy-based speech activity detector.

Speech rate: In computing speech-rate measures, we use a phone recognition algorithm based on a Hidden Markov Model approach, reported to have approximately an 80% phone-recognition accuracy [11]. We compute the number of speech units per second over the duration of a speech sample. We use the term *speaking rate* to refer to the phone rate over the total session time, with times when the speech is not active (pauses) included in the total session time. This is in contrast to *articulation rate*, which we compute as the phone rate over only the time during which speech is active. We also compute analogous rates using the concept of pseudosyllables derived from phone sequences [11].

4.0 CORRELATIONS WITH TOTAL ASSESSMENTS

Using the HAM-D and QIDS scales, in this section we calculate the correlation coefficient and p -value [15] between the vocal features of each of our three speech-production categories and the total scores of the depression scales. In the following section, we examine the correlation between the features and the individual symptoms of depression comprising the total scores. We have selected the Spearman correlation for our analysis due to a possible nonlinear relationship between the features and depression metrics [15]. The range of Spearman correlation magnitudes in this paper with $p < 0.05$ ($|r| \sim 0.10$ - 0.30), derived from average statistics, is consistent with previous measurements by Mundt et al. for the same database [3]. The clinical relevance of these correlations as indices of depression severity remains an open question for further research.

4.1 Source

Figure 2 illustrates correlation results and corresponding p values for our three source features against the HAM-D and QIDS total scores. As the algorithms used for jitter, shimmer, and HNR estimation assume stationarity, only held vowels from the Mundt database were analyzed. Consequently, we do not compare free and read speech for this case. Since there were four vowels and 28 subjects over four visits, there were 448 feature-and-score pairs used to calculate each of the correlations.

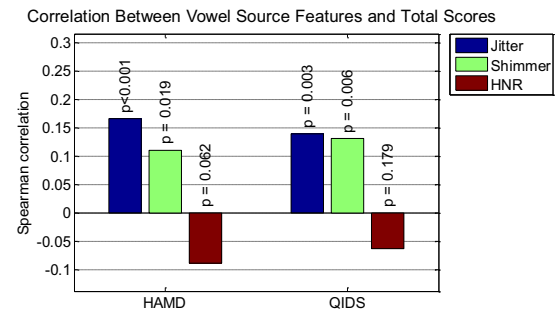


Figure 2: Correlations of vocal source features based on jitter, shimmer, and HNR against HAM-D and QIDS. The abscissa represents the Spearman correlation; the p -values are above each bar.

Although correlations are low, positive correlations associated with jitter and shimmer imply an increase in these variables, and thus what might be perceived as an increase in “hoarseness” with an increase in depression severity. Negative correlations for HNR imply an increase in aspiration with depression.

4.2 System

Figure 2 gives correlations of first and third formant features with the total score of HAM-D and QIDS for both free-response speech and the read Grandfather passage. The second formant (not shown), consistently showed negligible correlation. Except for average formant acceleration, correlations for all formant features with $p < 0.05$ are negative, implying a decline in formant dynamics with an increase in depression severity, consistent with what might be perceived as slurring in the depressed voice.

We also see in Figure 2 a comparison between free response and read speech, where negative correlations for the 1st and 3rd formants are generally larger for the Grandfather passage. Read speech may be a more sensitive probe of slurring in MDD than free-response speech because it involves less personal engagement, and is therefore more dependent on motivation or drive. The consistency of the Grandfather passage across subjects may also explain its stronger correlations.

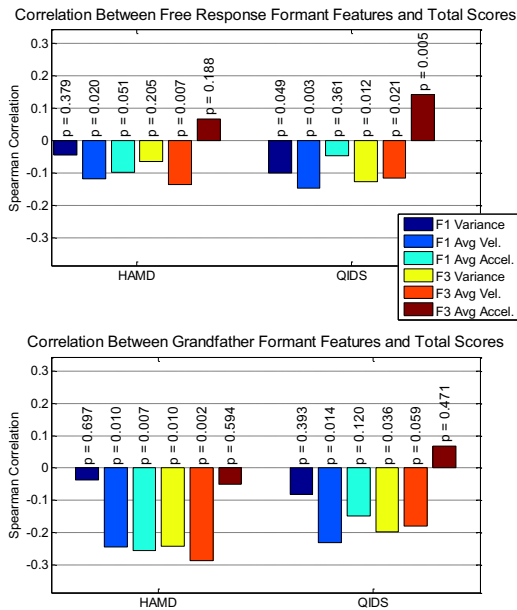


Figure 3: Correlations of formant variance and mean of formant velocity and acceleration against HAM-D and QIDS.

4.3 Prosody

Figures 4, 5, and 6 show correlations for our three prosodic features against the total HAM-D and QIDS depression scales for both free-response and read speech. From the figures, we observe the correlation differences between free-response and read speech, as well as between the two assessments.

For the energy-based features, negative correlations and corresponding p values of Figure 4 imply reduced energy dynamics with increasing depression severity, with read speech showing weaker negative correlations. For pitch-based

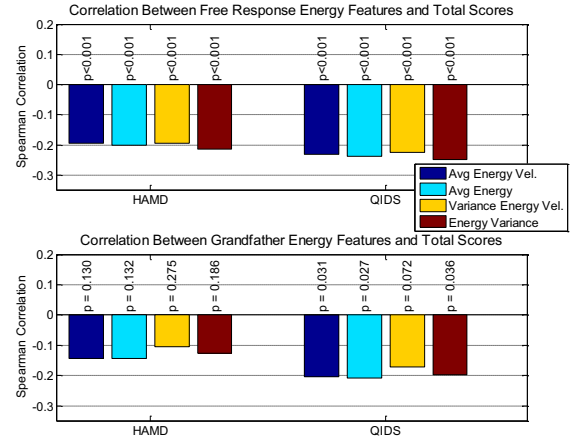


Figure 4: Correlations of average energy and velocity, and variance of energy and its velocity, against HAM-D and QIDS.

features, on the other hand, positive correlations of Figure 5 imply for free-response speech, and for QIDS assessment only, a more dynamic pitch contour with increasing depression severity, with read speech showing negligible correlations.

With our speech-rate features (Figure 6), in almost all cases correlations are negative, implying a slower rate with increasing depression severity. For read speech, there is a marked increase in the magnitude of these negative correlations, implying a slower rate of speaking in the reading state.

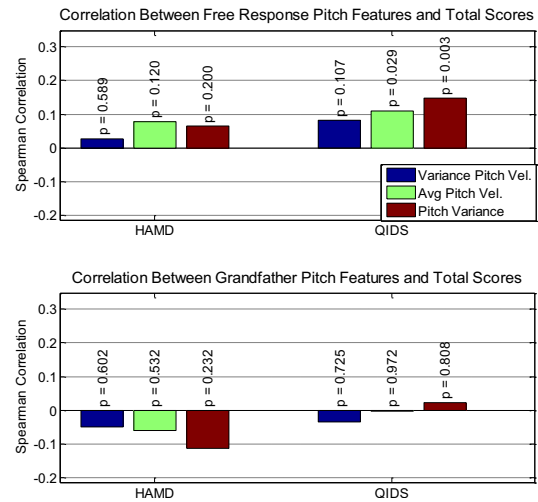


Figure 5: Correlations of average pitch velocity, and variance of pitch and its velocity, against HAM-D and QIDS assessments.

5.0 CORRELATION WITH SYMPTOM ASSESSMENTS

As described earlier, clinical and self-reported assessments of the level of MDD are obtained by adding the ratings of individual symptoms of depression; for the HAM-D and QIDS these symptoms are shown in Table 1.

As an extension of our initial study, we investigated correlations of our source, system, and prosody features with the individual symptoms. Correlations between component vocal features with individual symptoms of depression may reveal clearer relationships and dependencies that are not evident in the analysis of the total scale scores. There is a multitude of possibilities to consider. We present, as a representative example, correlations between each of the

features (except for the formant 2 features, which were found to be consistently negligible) and the individual HAM-D

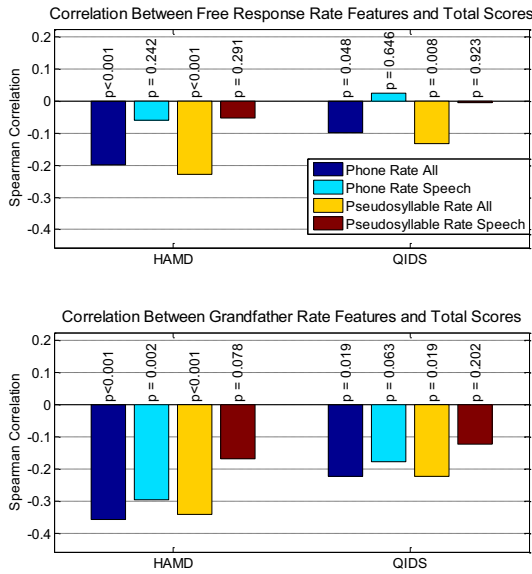


Figure 6: Correlations of vocal speech rate features based on speaking rate (with pauses) and articulation rate (without pauses), against Clinical HAM-D and QIDS. Rate parameters are derived from both phoneme and pseudo-syllable rates.

rating of depressed mood (feelings of sadness, hopelessness, helplessness, worthlessness) [20], shown in Figure 7.

It is evident that there is a range of correlations, and that a relatively small feature set is positively correlated (e.g., pitch measures, consistent with correlations with total assessment), while most are negatively correlated with depressed mood. It is also seen that in comparing results from free-response speech and the Grandfather passage, features are generally more negatively correlated with the latter, which is largely (but not always) consistent with the results obtained with the total scores. Although these observations have been made on the symptom of depressed mood, similar but symptom-specific observations can be made with other individual symptoms that are characteristic of MDD.

We have also observed that it is common for the correlation between a specific symptom and a feature to be greater than the correlation between the total score and that same feature. An example is shown in Figure 7 for the mean of velocity of formant 1, measured from both the free-response speech and the read Grandfather passage. As with correlations of individual symptoms with multiple features, the correlations of one feature with different depressive symptoms can be positive or negative, though there is a dominance of negative correlations. One outstanding case is psychomotor agitation (restlessness) [20] whose positive correlation indicates an increase in the dynamics of 1st formant velocity with increasing restlessness. As in Figure 7, features are generally more negatively correlated with read speech, which is consistent with the results obtained from the total scores.

6.0 CONCLUSIONS AND FUTURE WORK

In this paper, we investigated how source, system, and prosodic vocal features correlate with MDD using a database of voice recordings from 28 depressed subjects treated over a 6-week period [3]. Toward biomarkers for audio-based “on-body” continuous monitoring of depression severity, we

explored the contextual dependence of these correlations with free-response and read speech. We also investigated the

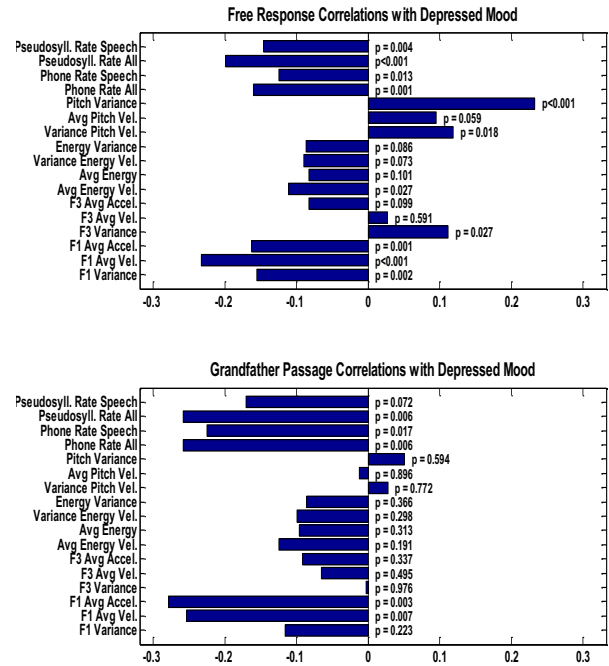


Figure 7: Spearman correlations of each of the features with the “depressed mood” sub-symptom of the HAM-D scale. Top panel: Spearman correlations for the free-response passages. Bottom panel: Spearman correlations for the grandfather passage. Abscissa in both axes is the Spearman correlation

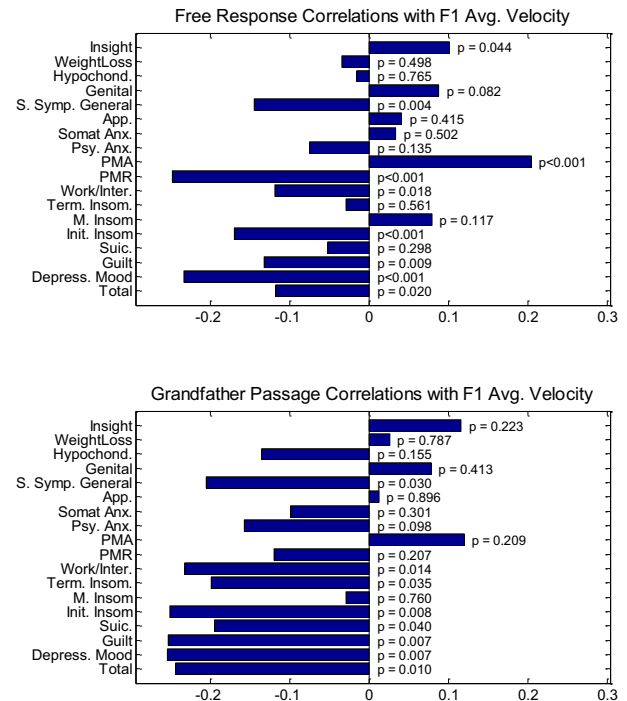


Figure 8: Spearman correlations of each of the individual MDD symptoms with the standard deviation of F1 velocity. Top panel: Spearman correlations for the free response. Bottom panel: Spearman correlations for the Grandfather Passage. Abscissa in both axes is the Spearman correlation

context- and speech component-dependence of correlations between our vocal features and assessment of individual

symptoms of MDD (e.g., depressed mood, agitation, energy). The predominant negative correlations, one exception being pitch in free-response speech, indicate a flatter affect with depression severity that may be more evident with our read Grandfather passage than with free-response speech.

Although the investigation presented here is based on a small dataset (28 subjects), the correlations found in the source, system, and prosodic features, and their tradeoffs with free-response and read speech, provide a foundation for our longer-term objective of “on-body” continuous monitoring in different life scenarios. For example, an implication of our results for future work in continuous monitoring for automatic classifying of depression state is that feature and symptom usage and their fusion must be evaluated in light of the speech-production context. This type of monitoring may provide continuous and unobtrusive measures of depressive states helpful in evaluating novel, faster-acting treatments. This type of technology might also be particularly helpful for monitoring rapidly changing psychiatric disorders, for example, manic bipolar disorder, or anxiety disorders such as post-traumatic stress disorder.

The contexts in this paper were limited (free-response vs. read speech) but other contexts that would be encountered in continuous monitoring, such as stress and personal dialogue scenarios, will likely show even stronger contextual dependencies. In addition, the type of analyses shown here, with objective, biologically-based performance measures for different contexts and symptoms, may further motivate movement away from subjective, clinical interviews assessing the full spectrum of diagnostic criteria toward more clearly defined symptom-based clusters reflective of underlying neurophysiologic pathology. Such a symptom-based approach is being championed by the National Institute of Mental Health under their Research Domain Criteria (RDoC) project. [25].

Finally, it is important to note that both the array of speech features we have selected, as well as their manner of computation, need to be expanded and refined in future work, and that the initial correlation analysis of this paper is currently serving as a guide to our on-going work in designing automatic classifiers of depression severity.

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